

Titanic Dataset - Exploratory Data Analysis

This report presents an Exploratory Data Analysis (EDA) of the Titanic dataset, focusing on patterns, trends, and relationships within the data.

```
In [ ]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style='whitegrid')
```

```
In [ ]: train_df = pd.read_csv('/mnt/data/train.csv')
```

Basic Information and Summary Statistics

```
In [ ]: train_df.info()
```

```
In [ ]: train_df.describe(include='all')
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column          Non-Null Count  Dtype
---  -
0   PassengerId     891 non-null   int64
1   Survived        891 non-null   int64
2   Pclass         891 non-null   int64
3   Name            891 non-null   object
4   Sex             891 non-null   object
5   Age            714 non-null   float64
6   SibSp          891 non-null   int64
7   Parch          891 non-null   int64
8   Ticket         891 non-null   object
9   Fare           891 non-null   float64
10  Cabin          204 non-null   object
11  Embarked       889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
count	891.000000	891.000000	891.000000	891	891	714.000000	891.000000	891.000000	891	891.000000	204	889
unique	NaN	NaN	NaN	891	2	NaN	NaN	NaN	681	NaN	147	3
top	NaN	NaN	NaN	Braund, Mr. Owen Harris	male	NaN	NaN	NaN	347082	NaN	B96 B98	S
freq	NaN	NaN	NaN	1	577	NaN	NaN	NaN	7	NaN	4	644
mean	446.000000	0.383838	2.308642	NaN	NaN	29.699118	0.523008	0.381594	NaN	32.204208	NaN	NaN
std	257.353842	0.486592	0.836071	NaN	NaN	14.526497	1.102743	0.806057	NaN	49.693429	NaN	NaN
min	1.000000	0.000000	1.000000	NaN	NaN	0.420000	0.000000	0.000000	NaN	0.000000	NaN	NaN
25%	223.500000	0.000000	2.000000	NaN	NaN	20.125000	0.000000	0.000000	NaN	7.910400	NaN	NaN
50%	446.000000	0.000000	3.000000	NaN	NaN	28.000000	0.000000	0.000000	NaN	14.454200	NaN	NaN
75%	668.500000	1.000000	3.000000	NaN	NaN	38.000000	1.000000	0.000000	NaN	31.000000	NaN	NaN
max	891.000000	1.000000	3.000000	NaN	NaN	80.000000	8.000000	6.000000	NaN	512.329200	NaN	NaN

Missing Values Check

```
In [ ]: train_df.isnull().sum()
```

We can observe missing values in the 'Age', 'Cabin', and 'Embarked' columns.

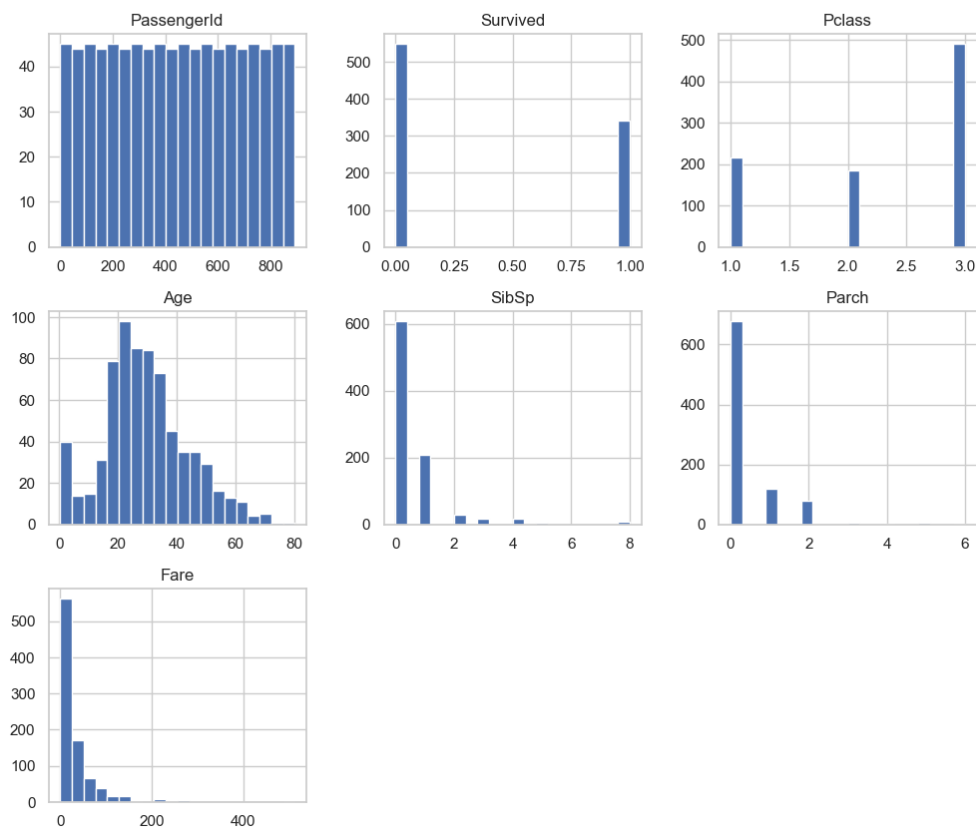
```
PassengerId      0
Survived          0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch           0
Ticket           0
Fare            0
Cabin          687
Embarked         2
dtype: int64
```

Distributions of Numerical Features

```
In [ ]: train_df.hist(figsize=(12, 10), bins=20)
plt.suptitle("Distributions of Numeric Columns", fontsize=16)
plt.show()
```

Most numerical features like 'Fare' and 'Age' show skewed distributions, indicating the presence of outliers or non-normal behavior.

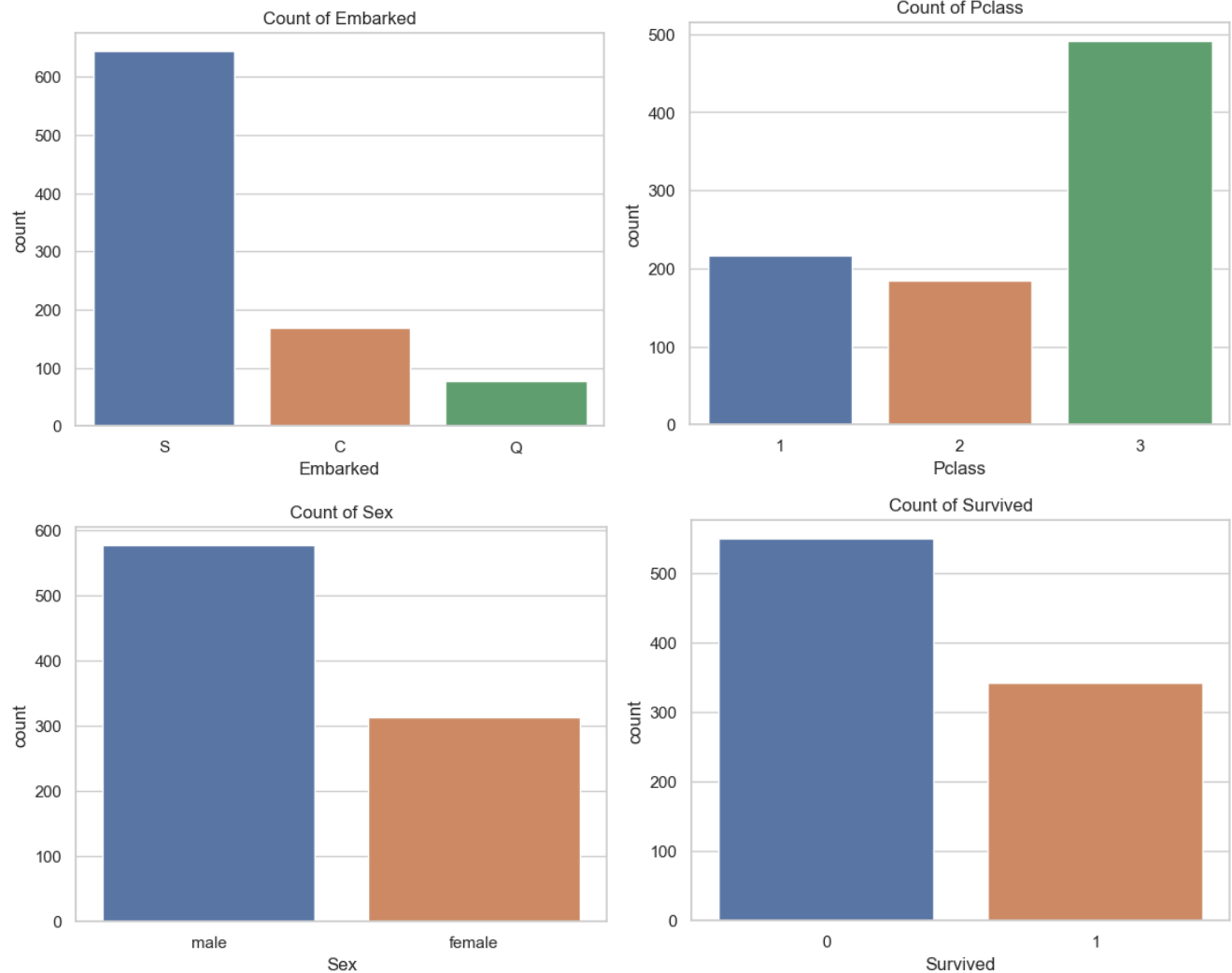
Distributions of Numeric Columns



Count of Categorical Variables

```
In [ ]: categorical_cols = ['Survived', 'Pclass', 'Sex', 'Embarked']
for col in categorical_cols:
    sns.countplot(x=col, data=train_df)
    plt.title(f"Count of {col}")
    plt.show()
```

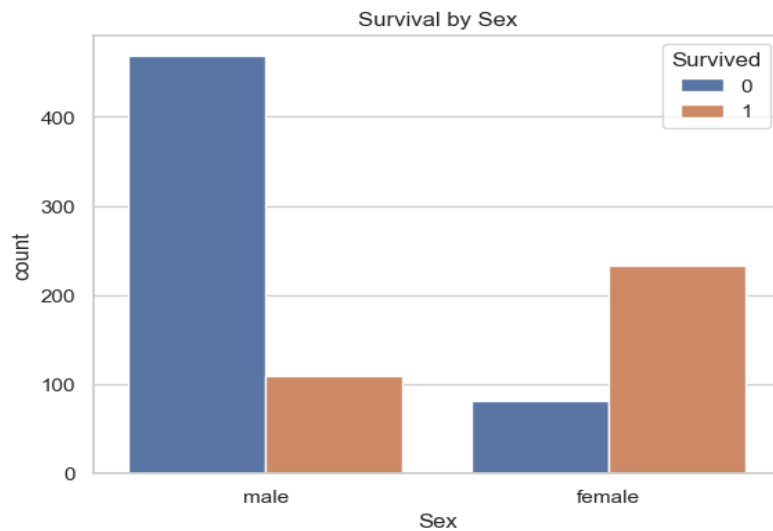
The dataset contains more passengers in 3rd class, and more males than females.



Survival by Sex

```
In [ ]: sns.countplot(x='Sex', hue='Survived', data=train_df)
pH.title("Survival by Sex")
pH.show()
```

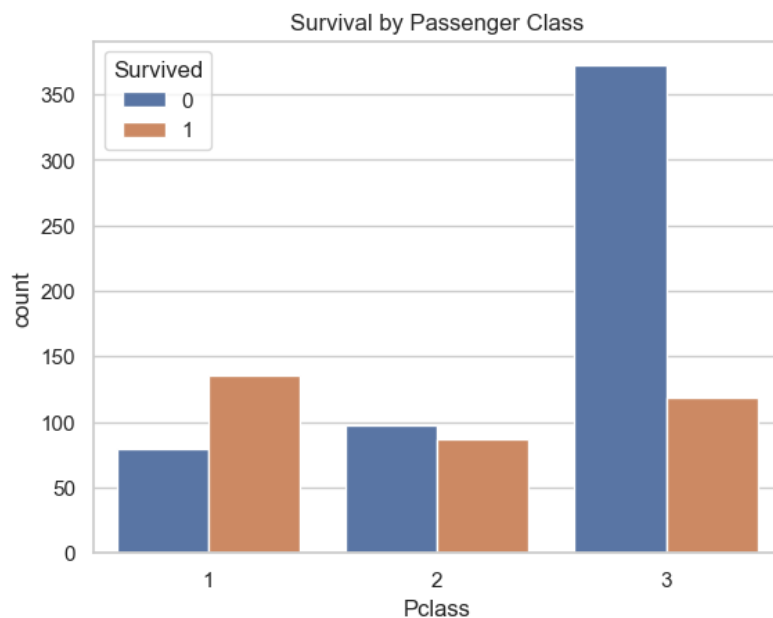
A significantly higher proportion of females survived compared to males.



Survival by Passenger Class

```
In [ ]: sns.countplot(x='Pclass', hue='Survived', data=train_df)
pH.title("Survival by Passenger Class")
pH.show()
```

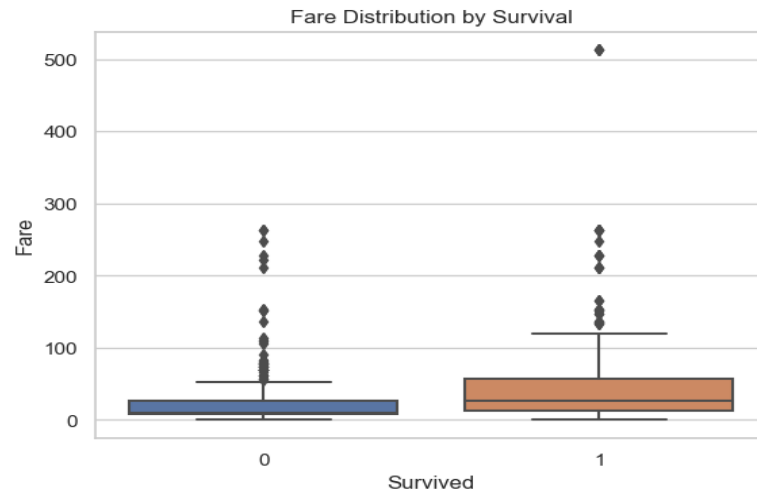
First class passengers had higher survival rates, indicating socio-economic factors may have played a role.



Fare Distribution by Survival

```
In [ ]: sns.boxplot(x='Survived', y='Fare', data=train_df)
pH.title("Fare Distribution by Survival")
pH.show()
```

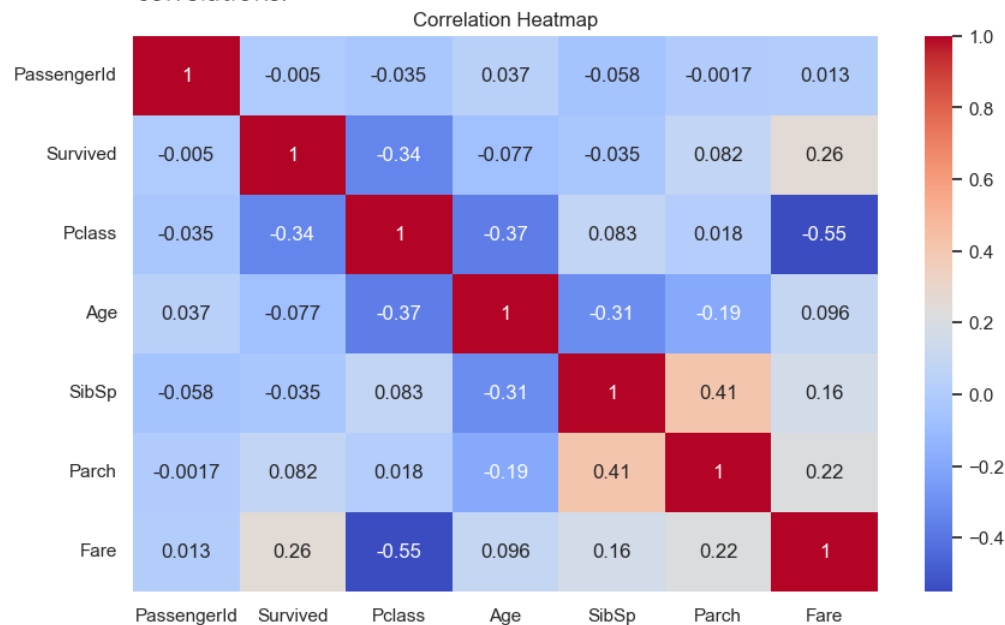
Survivors tended to have paid higher fares on average.



Correlation Heatmap

```
In [ ]: plt.figure(figsize=(10,6))
sns.heatmap(train_df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
pH.show()
```

Fare and Pclass show correlation with Survival. Parch and SibSp also show slight positive correlations.

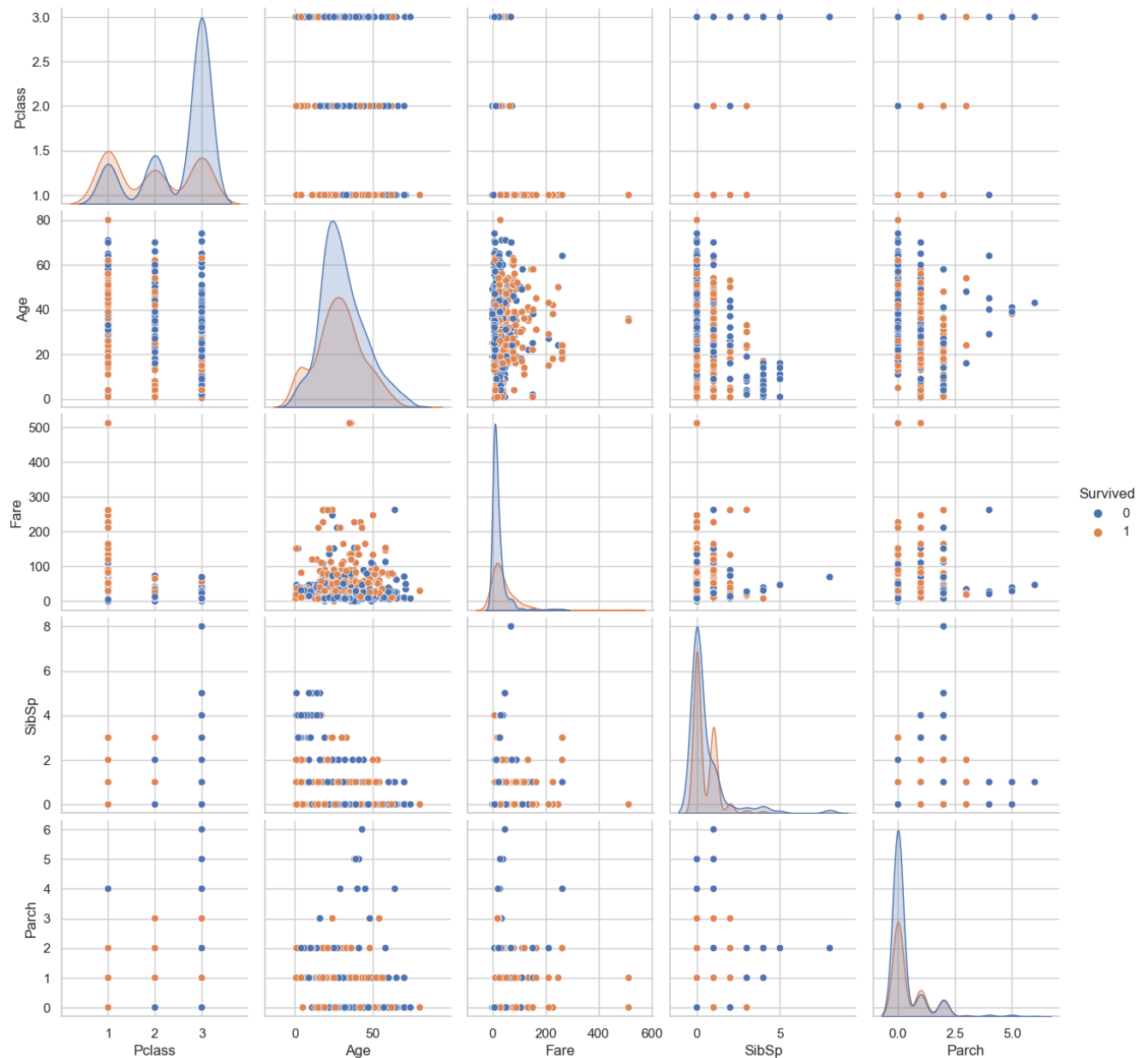


Pairwise Relationships

```
In [ ]: selected_features = ['Survived', 'Pclass', 'Age', 'Fare', 'SibSp', 'Parch']
sns.pairplot(train_df[selected_features], hue='Survived')
pH.suptitle("Pairwise Relationships", y=1.02)
pH.show()
```

Interesting clustering is visible for features like Fare and Pclass when compared to survival.

Pairwise Relationships



Conclusion

- Females and first-class passengers had higher survival rates.
- Higher fares were associated with better survival chances.
- Socioeconomic status, age, and gender had a clear impact on survival.